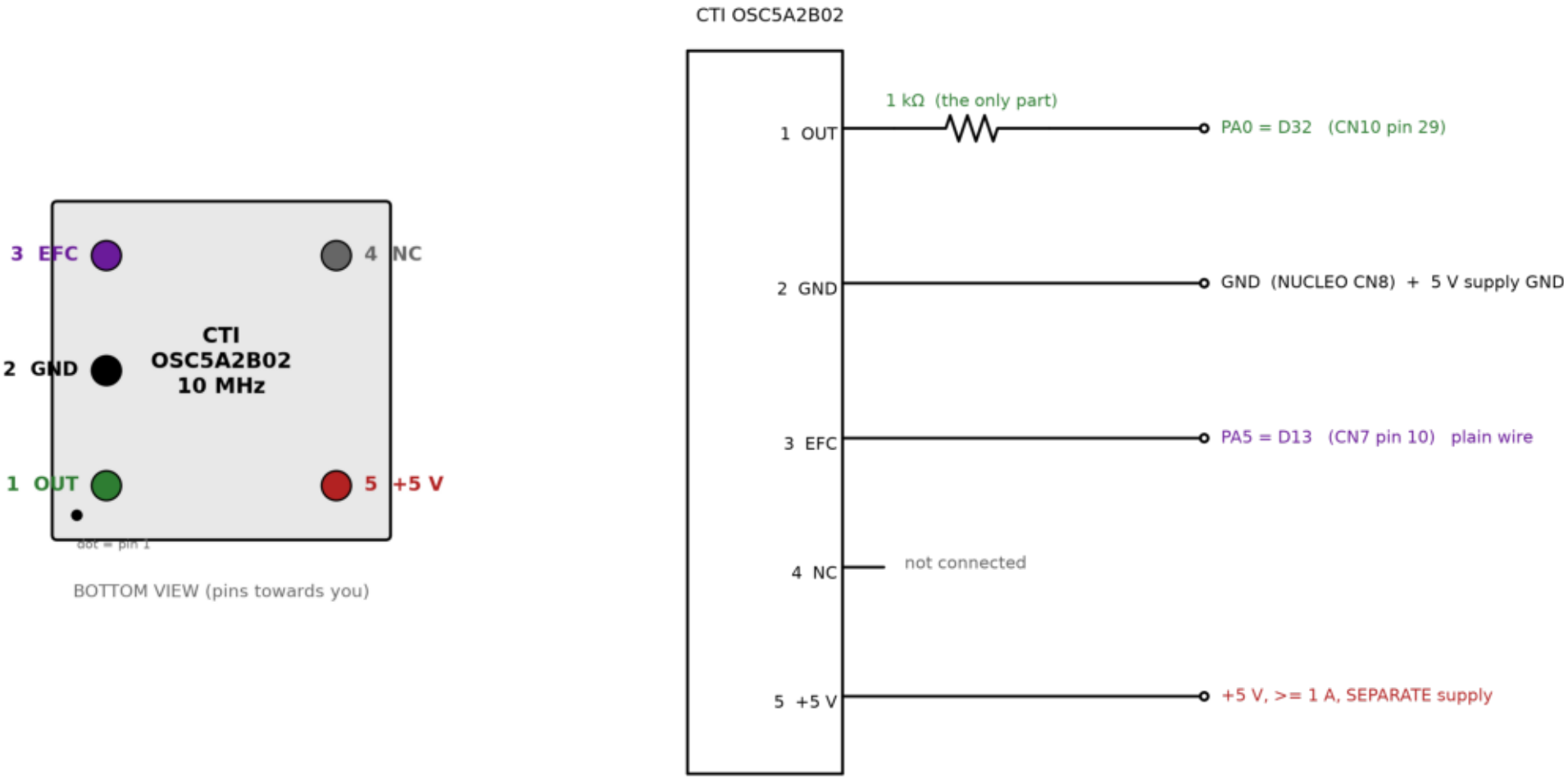
# OCXO test, variant B (minimal): 4 wires + one 1 kΩ resistor

CTI OSC5A2B02 -> NUCLEO-H723ZG. Firmware build -DOCXO_REF (commit 9eab67b): TIM2 counts the OCXO on PA0, the DAC on PA5 drives its EFC.




1. Power ORDER: NUCLEO powered first (USB), then the OCXO's 5 V supply; switch off in the reverse order.
2. Pin 5 -> +5 V from a separate supply (>= 1 A: 600 mA for the first minutes, 250 mA warm). Pin 2 -> that supply's GND AND NUCLEO GND.
3. Pin 1 -> 1 kΩ -> PA0 / D32. The pin's own protection clamps the 5 V square at 3.3 V; 1 kΩ limits that current to about 1.5 mA (allowed: 5 mA). Keep this wire short (< 10 cm) and twist it with the ground wire. An 0805 on a jumper wire + heat-shrink is fine.
4. Pin 3 (EFC, marked 'Vref' on the datasheet) -> PA5 / D13 with a plain wire. The DAC starts at mid-scale = 1.65 V (about -0.3 ppm, harmless); 0 ... 3.3 V covers about -1.5 ... +0.9 ppm. Pin 4: nothing.
5. Let the OCXO warm up >= 10 minutes, then the OCXO firmware is flashed. 'OCXO reference OK' in the log = 4 kHz captures found.